FARAAZ MOHAMMED

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Full Australian working rights (Temporary Graduate visa, subclass 485) — no sponsorship required • Available immediately

PROFILE

Master of Artificial Intelligence graduate (RMIT, 2026) whose AI/ML engineering internship centred on evaluation: IoU and sensitivity harnesses, a satellite-verified ground-truth set, and a published negative result. Builds applied LLM systems end to end in Python: a cost-proportional Claude model router, a fine-tuning loop that turns user feedback into a QLoRA dataset, and retrieval-grounded generation. Lets data settle arguments. Seeking a Machine Learning Engineer (Applied LLMs) role; open to relocating to Sydney.

TECHNICAL SKILLS

Languages: Python (primary, production: FastAPI, SQLAlchemy, pandas), SQL, TypeScript, JavaScript, Rust, Bash

LLM systems: evaluation design (precision/recall, IoU, sensitivity analysis, ground-truth construction), retrieval-grounded generation, LLM fine-tuning (Unsloth, QLoRA, GGUF), Ollama, Anthropic API (prompt caching, routing), MCP servers

Applied ML: scikit-learn, XGBoost, OpenCV, pandas, GeoPandas / Shapely, Kestra

Platforms & tooling: Git / GitHub, Docker, AWS (CloudFormation, Fargate, RDS, CloudFront), PostgreSQL, Colab, Linux, Claude Code (skills, hooks, subagents), Cursor

EXPERIENCE

Full-Stack Developer Intern — Vayaa
Sep 2026 – present

Melbourne, VIC · hybrid · company website: Python (FastAPI) + PostgreSQL, React + TypeScript, deployed on AWS

- Built the contact-enquiry API in Rust, then led its port to Python (FastAPI) on the mentor's call, gating the cutover on 50 parity tests passing against both backends.
- Took it live on AWS with CloudFormation (CloudFront → ECS Fargate → RDS PostgreSQL), fixing Free-plan limits, a CloudFront account block and RDS TLS (verify-full); kept a decision log (choice, alternatives, cost) for every call.

Tools: Python, FastAPI, SQLAlchemy, PostgreSQL, Docker, AWS, Rust (Axum), React, TypeScript

AI/ML Engineering Intern — Prodegee
Mar 2026 – Jun 2026

Remote, Melbourne · RMIT industry capstone, Team DS4 — building-footprint inference for off-grid electrification planning

- Built an IoU evaluation harness for compound polygons and traced all three generators' failures to a ~35× over-scoped upstream cluster rather than a fitting fault, which redirected the next assignment.
- Built a ±20% sensitivity harness that identified cluster radius as the dominant parameter (±36–44% swing across 191 villages), driving a 75 → 200 m recalibration that fixed a 90%+ household under-count (0.069 → 0.415).
- Co-built a 40-school satellite-verified ground-truth set and used it to correct a published baseline (6–13% → 37%); published a negative result showing a proximity penalty hurt accuracy at every threshold.
- Trained an XGBoost multi-output tier regressor over 200 villages in 7 Mauritanian Wilayas; documented that its 0.816 "confidence" measured agreement with the training heuristic, not accuracy.
- Designed and built the school-geolocation algorithm at the core of the team's pipeline (7.2M Overture building polygons, BFS spatial clustering, composite scoring); ran it on 18,121 schools in Côte d'Ivoire and ported it to Burundi and Chad.
- Replaced a binary distance cutoff with exponential-decay scoring and a weighted confidence score; added per-country sigmoid normalisation so one threshold works in all three countries.
- Built a Claude API model router that classifies task complexity with a cheap Haiku call and routes to Haiku / Sonnet / Opus, keeping cost proportional to difficulty.
- Led 3 of 10 team assignments and was primary author of 3 of 4 formal deliverables (68 commits, 7 peer reviews); pinned upstream inputs by commit SHA for reproducible re-runs.

Tools: Python, scikit-learn, XGBoost, pandas, GeoPandas, Shapely, Kestra, Anthropic API, Git, Claude Code (MCP, hooks)

Dispatch Team Member — Amart Furniture
Feb 2025 – Mar 2026

Melbourne, VIC · part-time, on-site, concurrent with full-time study

- Tracked and reconciled stock in the inventory system, processed customer orders and resolved fulfilment discrepancies across 5 Melbourne stores; trained a new team member.

KEY PROJECTS

Gemma Delegation System — local/cloud LLM hybrid
Node.js, MCP SDK, Ollama, Unsloth/QLoRA, Colab — *May 2026*

- Built an MCP server that lets Claude Code or Cursor delegate lightweight tasks to a locally hosted Gemma model instead of spending frontier-model tokens, failing open to the frontier model whenever Ollama is unavailable.
- Closed the loop: accept/reject feedback updates a capability profile and curates a JSONL dataset from real usage; a Colab notebook fine-tunes on confirmed examples (4-bit QLoRA, LoRA r=16) and exports GGUF back into Ollama.

↳ GitHub: github.com/faraaz817/gemma-delegation-system

Bro — interactive AI portfolio chatbot · Next.js, TypeScript, Prisma/Postgres, Anthropic API *2026*

- Built a recruiter-facing site that answers questions from a profile corpus via server-side retrieval with anti-hallucination constraints; deployed on Vercel.
  ↳ Live: job-go-mauve.vercel.app/portfolio

Agent skills for Claude Code / Cursor · skills, hooks, reusable workflows *2026*

- Authored five reusable, public agent skills — CO-STAR requirements builder, Socratic tutor, better-way reviewer, skill sync, and a pikepdf CLI that unlocks signature-locked PDFs.
  ↳ GitHub: github.com/faraaz817/My_Skills
- **Fire Detection & Alerting System (B.Tech capstone, team of 4):** trained a custom OpenCV Haar cascade (Python) for real-time fire detection, chosen over heavier detectors for latency; a threaded pipeline raises an alarm and sends WhatsApp alerts to the nearest fire station.

EDUCATION

Master of Artificial Intelligence — RMIT University *Jul 2024 – Jul 2026*
Melbourne, VIC · Machine Learning, Computer Vision, AI Systems Design, Intelligent Decision Making (ASP/Clingo), IoT

Bachelor of Technology (Information Technology) — Keshav Memorial Institute of Technology *Dec 2020 – May 2024*
Hyderabad, India (affiliated to JNTUH) · Coursework incl. Machine Learning, Neural Networks & Deep Learning, Data Mining

PUBLICATION

K. M. Asudaria, **M. F. Abdul Khadeer**, S. S. Rafai, K. S. Reddy. "A Comparative Study of Edge Detection Techniques to Identify Maize Leaf Diseases using Machine Learning." *Quest Journal of Software Engineering and Simulation*, Vol. 9(12), pp. 43–51, Dec 2023 (ISSN 2321-3795, peer-reviewed). Benchmarked 5 edge-detection operators × 8 classifiers (40 combinations) on a 15,000-image, four-class dataset; best result Sobel + MLP at 94.75%. ↳ Code: github.com/faraaz817/Machine_Learning

CERTIFICATIONS, AWARDS & LEADERSHIP

- Smart India Hackathon delegate, Government of India (2022) · Certificate of Publication, Quest Journals (2023).
- Head, Vachan Speakers Club, KMIT (2022–23) — led public-speaking workshops, debates and communication seminars.

LANGUAGES

English (native / bilingual) • Urdu (native / bilingual) • Hindi (full professional) • Telugu (elementary)

REFEREES

Available on request.